

MTH2021 Mathematical Methods 2

Complex Analysis – Optional Topics

There are many extensions to the material from lectures, which we do not have time for but which are practically important and link up with other modules in Applied Mathematics. These notes give a summary of some of these further developments.

1. Analytic functions

1.1. Cauchy-Riemann (CR) equations in polar form

The most familiar form of the Cauchy-Riemann (CR) equations is in terms of the Cartesian variables x, y in $z = x + iy$:

$$u_x = v_y \quad \text{and} \quad u_y = -v_x \quad (1)$$

where we have written an analytic function $f(z)$ in terms of its real and imaginary parts u, v as

$$w = f(z) = u(x, y) + iv(x, y). \quad (2)$$

We occasionally use the short-hand notation u_x for the partial derivative $(\partial u / \partial x)_y$ etc, when it is clear from the context what is kept constant in taking the derivative. We will use the full notation in variable transformations and the chain rule, to keep track of the variables involved.

We can also derive a version of the CR equation in terms of the plane polar coordinates r, θ in $z = r \exp(i\theta)$. To this end, we write

$$w = f(z) = u(r, \theta) + iv(r, \theta). \quad (3)$$

We pick a point z and consider two different approaches to it: along the ray $\theta = \text{const}$ and along the circle $r = \text{const}$. Then

$$\Delta z = \Delta r \exp(i\theta) \quad \text{along } \theta = \text{const} \quad (4)$$

$$\Delta z = ir \Delta \theta \exp(i\theta) \quad \text{along } r = \text{const}. \quad (5)$$

By the definition of differentiability, we require

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{\Delta w}{\Delta z} \quad (6)$$

to be the same along the two directions. Writing $\Delta w = \Delta u + i\Delta v$ then gives

$$u_r + iv_r = \frac{1}{ir} (u_\theta + iv_\theta). \quad (7)$$

Equating real and imaginary parts gives the CR conditions in polar form:

$$ru_r = v_\theta \quad \text{and} \quad rv_r = -u_\theta. \quad (8)$$

This form of the CR conditions can be very handy. For example, consider $f(z) = z^n$:

$$f(z) = z^n = r^n \exp(in\theta) = r^n \cos(n\theta) + ir^n \sin(n\theta) \quad (9)$$

$$\Rightarrow u = r^n \cos(n\theta), \quad v = r^n \sin(n\theta). \quad (10)$$

Hence

$$ru_r = nr^n \cos(n\theta) \quad , \quad v_\theta = nr^n \cos(n\theta) \quad (11)$$

$$rv_r = nr^n \sin(n\theta) \quad , \quad u_\theta = -nr^n \sin(n\theta) \quad (12)$$

showing that the CR conditions are fulfilled and the function is analytic. The only place where the polar form of the CR conditions runs into trouble is $z = 0$, where $r = 0$ and θ is undefined. In the above example, if $n < 0$, then $f(z)$ is not analytic at $z = 0$ but we won't glean that from the CR equations in polar form, as we cannot apply them at $z = 0$.

Another example:

$$f(z) = (z/z^*)^n = \exp(2in\theta) = \cos(2n\theta) + i\sin(2n\theta) \quad (13)$$

$$\Rightarrow u = \cos(2n\theta) \quad , \quad v = \sin(2n\theta) . \quad (14)$$

Then

$$ru_r = 0 \quad , \quad v_\theta = 2n \cos(2n\theta) \quad (15)$$

$$rv_r = 0 \quad , \quad u_\theta = -2n \sin(2n\theta) . \quad (16)$$

The CR equations are not obeyed at any $z \neq 0$ and the function is not analytic anywhere. The function is furthermore not defined at $z = 0$, but the fact that it is not differentiable anywhere else is enough to conclude that it is nowhere analytic (since a function analytic at a point z is, by definition, differentiable at z and throughout a neighbourhood of z).

1.2. The “ z^* test”

The CR equations have an interesting consequence. Step back to Cartesian variables and write a function $f(z)$ as in (2): $f(z) = u(x, y) + iv(x, y)$. Here x and y are real. But let us now extend x and y to be two independent complex variables. This enables us to treat

$$z = x + iy \quad \text{and} \quad \bar{z} = x - iy \quad (17)$$

as another pair of independent complex variables. In the special case when x and y are real, $\bar{z}$ reduces to $\bar{z} = z^*$. We can also express x and y in terms of z and $\bar{z}$:

$$x = \frac{z + \bar{z}}{2} \quad \text{and} \quad y = \frac{z - \bar{z}}{2i} . \quad (18)$$

f depends on x and y via $u = u(x, y)$ and $v = v(x, y)$, and x and y are functions of z and $\bar{z}$. Thus, we can think of f as being a function of z and $\bar{z}$:

$$f = f(z, \bar{z}) . \quad (19)$$

We now embark on an exercise in partial derivatives:

$$\left(\frac{\partial f}{\partial \bar{z}} \right)_z = \left(\frac{\partial f}{\partial x} \right)_y \left(\frac{\partial x}{\partial \bar{z}} \right)_z + \left(\frac{\partial f}{\partial y} \right)_x \left(\frac{\partial y}{\partial \bar{z}} \right)_z \quad (20)$$

$$= \frac{1}{2} \left[\left(\frac{\partial f}{\partial x} \right)_y + i \left(\frac{\partial f}{\partial y} \right)_x \right] \quad (21)$$

$$= \frac{1}{2} \left\{ \left[\left(\frac{\partial u}{\partial x} \right)_y - \left(\frac{\partial v}{\partial y} \right)_x \right] + i \left[\left(\frac{\partial v}{\partial x} \right)_y + \left(\frac{\partial u}{\partial y} \right)_x \right] \right\} . \quad (22)$$

If f is analytic, then u and v satisfy the CR conditions $u_x = v_y$, $u_y = -v_x$. Then both terms above vanish and

$$\left(\frac{\partial f}{\partial \bar{z}}\right)_z = 0. \quad (23)$$

Conclusion: if f , when written as a function of z and $\bar{z}$, does not obey (23), then it cannot be analytic. For example

$$f = x = \frac{z + \bar{z}}{2} \Rightarrow \left(\frac{\partial f}{\partial \bar{z}}\right)_z = \frac{1}{2} \neq 0 \quad (24)$$

and thus f is not analytic. Similarly, $f = x - iy = \bar{z}$ is not analytic, nor is $f = x^2 + y^2 = z\bar{z}$ etc.

Thus we reach yet another view of how analytic functions arise: they are expressible as functions of $z = x + iy$ alone, without “mixing in” the “rogue” variable $\bar{z} = x - iy$.

1.3. Reconstructing an analytic function from its real or imaginary part

Suppose someone hands us a function $g(x, y)$. Two questions we can ask are:

- i. can g be the real or imaginary part of an analytic function;
- ii. if so, how can we find this function?

To answer the first question we recall that the real and imaginary parts, u and v , of an analytic function individually satisfy Laplace’s equation: $u_{xx} + u_{yy} = 0$, $v_{xx} + v_{yy} = 0$. The converse is also true: if $g(x, y)$ satisfies Laplace’s equation, then there exists an analytic function such that $g(x, y)$ is its real or imaginary part. Thus, we can answer i. by checking whether or not the given g solves Laplace’s equation.

Suppose it does. We now face ii.: finding the analytic function, of which g is the real or imaginary part. There are different ways of doing this. Here we consider a quick method, based on the earlier results. We recall that if $f(z) = u(x, y) + iv(x, y)$ is analytic, then we can write its derivative as

$$\frac{df}{dz} = u_x + iv_x = u_x - iu_y. \quad (25)$$

But

$$\left(\frac{\partial u}{\partial x}\right)_y = \left(\frac{\partial u}{\partial z}\right)_{\bar{z}} + \left(\frac{\partial u}{\partial \bar{z}}\right)_z \quad (26)$$

$$\left(\frac{\partial u}{\partial y}\right)_x = i \left(\frac{\partial u}{\partial z}\right)_{\bar{z}} - i \left(\frac{\partial u}{\partial \bar{z}}\right)_z. \quad (27)$$

Hence,

$$\frac{df}{dz} = 2 \left(\frac{\partial u}{\partial z}\right)_{\bar{z}} \quad (28)$$

and therefore

$$f(z) = 2u[x(z, \bar{z}), y(z, \bar{z})] + c(\bar{z}) \quad (29)$$

where the dependence of x and y on z and $\bar{z}$ is given in (18) above, and $c(\bar{z})$ is such as to make the right-hand side of (29) independent of $\bar{z}$ overall. Then we can set $\bar{z}$ to anything we choose, say $\bar{z} = 0$. Then (29) becomes

$$f(z) = 2u(z/2, z/2i) + c. \quad (30)$$

By considering $z = 0$ (where $f(0) = u(0, 0) + iv(0, 0)$) we see that $c = -f^*(0)$ so that

$$f(z) = 2u(z/2, z/2i) - f^*(0). \quad (31)$$

For example, suppose we have $u(x, y) = x^2 - y^2$, $f(0) = 0$. Then

$$f(z) = 2[(z/2)^2 - (z/2i)^2] = z^2. \quad (32)$$

By analogous steps, we can reconstruct f from its imaginary part, v :

$$f(z) = 2iv(z/2, z/2i) + f^*(0). \quad (33)$$

1.4. The “constant- f ” results

There is a family of results forcing an analytic function to be constant under certain conditions.

- i. If $f(z)$ is analytic and pure real or pure imaginary in a region R , then $f(z)$ is constant in R . If $f(z)$ is pure real, say, then $f = u + iv$, where $v = 0$. By the CR conditions then $u_x = v_y = 0$, $u_y = -v_x = 0$ and hence $u = f$ is constant.
- ii. If both $f(z)$ and $f^*(z)$ are analytic in R , then $f(z)$ is constant in R . If $f(z)$ and $f^*(z)$ are analytic, then so is $f(z) + f^*(z)$ (since the sum of two analytic functions is analytic). But $f(z) + f^*(z)$ is pure real, and is hence constant by i. above. Similarly, $f(z) - f^*(z)$ is constant. Hence $f(z)$ is constant.
- iii. If $f(z)$ is analytic and $|f(z)|$ is constant in R , then $f(z)$ is constant in R . If $|f(z)|$ is constant, then $f(z)f^*(z) = c = \text{const.}$ Hence $f^*(z) = c/f(z)$. But $c/f(z)$ is analytic, if $f(z)$ is. Hence $f^*(z)$ is analytic. Then $f(z)$ is constant by ii. above.

1.5. Geometrical meaning of the derivative

The derivative, $f'(z)$, of an analytic function $f(z)$ is a complex number with its own modulus and argument

$$|f'(z)| \quad \text{and} \quad \arg f'(z). \quad (34)$$

They have a beautiful geometrical meaning. We recall that a function $f(z)$ is a map from the complex plane where $z = x + iy$ lives, to a complex plane inhabited by $w = f(z) = u + iv$. Think of $f'(z)$ as the ratio of two small changes, Δz in z and Δw in w :

$$f'(z) = \frac{\Delta w}{\Delta z} \quad \Rightarrow \quad \Delta w = f'(z) \Delta z. \quad (35)$$

Therefore

$$|\Delta w| = |f'(z)| |\Delta z| \quad (36)$$

$$\arg \Delta w = \arg \Delta z + \arg f'(z). \quad (37)$$

We can now see pictorially what the map $w = f(z) = u + iv$ does. First, it takes small (infinitesimal) line segments and relocates them, from position (x, y) in the z -plane to position (u, v) in the w -plane. Second, it rescales them in length by a factor $|f'(z)|$; finally, it rotates them through an angle $\arg f'(z)$: the image of Δz , namely Δw , has been scaled and rotated relative to Δz .

1.6. Conformal maps

We can take the last observation a step further. The derivative, $f'(z) = \Delta w / \Delta z$, of an analytic function is independent of the direction from which $(z + \Delta z)$ approaches z , that is, of $\arg \Delta z$. Make the approach along two different curves, C_1 and C_2 , in the z -plane, intersecting at z under an angle α . Let Δz_1 and Δz_2 be the increments in z along the two directions, and let Δw_1 and Δw_2 be the respective changes in w . Then

$$f'(z) = \frac{\Delta w_1}{\Delta z_1} = \frac{\Delta w_2}{\Delta z_2} \quad (38)$$

$$\Rightarrow \frac{\Delta w_1}{\Delta w_2} = \frac{\Delta z_1}{\Delta z_2} \quad (39)$$

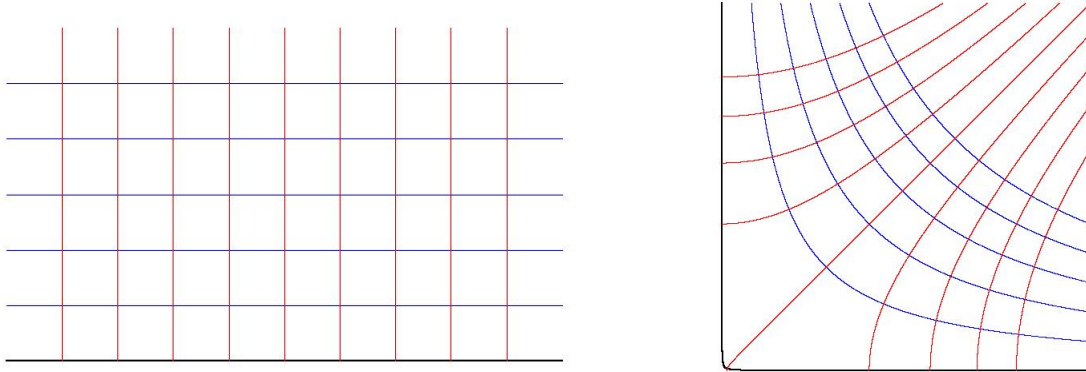
$$\Rightarrow \arg \left(\frac{\Delta w_1}{\Delta w_2} \right) = \arg \left(\frac{\Delta z_1}{\Delta z_2} \right) = \alpha. \quad (40)$$

Then the images of C_1 and C_2 in the w -plane intersect under the same angle (α) as that between C_1 and C_2 in the z -plane. Transformations that preserve angles of intersection of trajectories are called conformal transformations. Analytic functions are conformal transformations.

For example, consider the transformation

$$w = u + iv = f(z) = z^2 = (x^2 - y^2) + i(2xy). \quad (41)$$

It maps the upper half w -plane to the positive quadrant of the z -plane. Here are the families of curves $u = \text{const}$ and $v = \text{const}$ in the w -plane (left) and their images in the z -plane, i.e. the families of curves $x^2 - y^2 = \text{const}$ and $2xy = \text{const}$ (right):



As expected, the angle of intersection in the two planes is the same. But there is more to it. The angle in question, in this example, is $\pi/2$ (by construction), and thus in each plane we have an orthogonal family of curves. This would remain the case if we replaced $f(z) = z^2$ above by any analytic function: $u = \text{const}$ and $v = \text{const}$ are orthogonal in the w -plane in all cases, and, by (40), their images $u(x, y) = \text{const}$ and $v(x, y) = \text{const}$ in the z -plane would have to be orthogonal, always. This observation takes us to one of the great applications of analytic functions.

2. Solutions to Laplace's equation

2.1. Three for the price of one

We know from lectures that, as a direct consequence of the CR equations, the real and imaginary parts $u(x, y)$ and $v(x, y)$ of an analytic function $f = u + iv$ are harmonic functions, that is, they solve Laplace's equation:

$$u_{xx} + u_{yy} = 0 \quad \text{and} \quad v_{xx} + v_{yy} = 0. \quad (42)$$

Therefore, by picking any analytic function, we are automatically getting two solutions to Laplace's equation (in two dimensions) for free. Furthermore, using the CR equations:

$$\nabla u \cdot \nabla v = u_x v_x + u_y v_y \quad (43)$$

$$= u_x v_x + (-v_x) u_x = 0. \quad (44)$$

Thus, as expected from the end of the last section, $u(x, y) = \text{const}$ and $v(x, y) = \text{const}$ are orthogonal families. This third benefit completes the package. The pair of harmonic functions u and v are sometimes called harmonic conjugates. To see how useful this is, consider a physical example.

The electrostatic potential, call it V , in free space (away from charges) obeys Laplace's equation. Further, the electric field, $\mathbf{E} = -\nabla V$, is orthogonal to the equipotentials $V = \text{const}$. Thus, the real (or imaginary) part u (or v) of an analytic function $f = u + iv$ can represent the electrostatic potential in a suitable geometry, while the lines of constant v (or u) - the harmonic conjugate of the potential - give the electric-field lines. For instance, the blue (horizontal) lines in plot 1 above represent the equipotentials above a horizontal charged metal plate (extended along the axis perpendicular to the page), with the red (vertical) lines being the field lines. In plot 2, blue are the equipotentials inside two charged metal plates meeting at right angles, and red are the field lines. By considering $f(z) = z^n$, we can generate the potential and field lines inside pairs of plates meeting at other angles. And so on.

2.2. Fluid flow

Laplace's equation arises in numerous contexts. Another example is fluid flow. Let ρ be the density of the fluid and $\mathbf{v}$ - the velocity of the flow. If the flow is irrotational, then

$$\nabla \times \mathbf{v} = 0 \quad \Rightarrow \quad \mathbf{v} = \nabla \Phi \quad (45)$$

where Φ is the velocity potential. If the flow is, further, incompressible ($\rho = \text{const}$ in space and in time), then the continuity equation (which ensures mass conservation) $\partial \rho / \partial t + \nabla \cdot (\rho \mathbf{v}) = 0$ gives

$$\nabla \cdot \mathbf{v} = 0 \quad \Rightarrow \quad \nabla^2 \Phi = 0. \quad (46)$$

Therefore, the velocity of the flow $\mathbf{v}$ is deriveable as the gradient of a harmonic function, Φ .

For example, consider

$$f(z) = z + \frac{1}{z}. \quad (47)$$

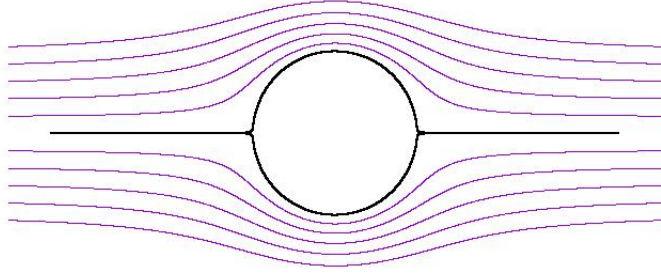
Writing z in polar form, $z = r \exp(i\theta)$:

$$f(z) = \left(r + \frac{1}{r}\right) \cos \theta + i \left(r - \frac{1}{r}\right) \sin \theta = \Phi + i\Psi. \quad (48)$$

Therefore

$$\Psi = \left(r - \frac{1}{r} \right) \sin \theta \quad (49)$$

is constant (and equal to zero) on the x -axis ($\theta = 0, \theta = \pi$) together with the unit circle ($r = 1$). The family of curves $\Psi = \text{const}$ can represent the streamlines for ideal flow past a cylinder centered at the origin (and extended perpendicular to the page):



The velocity potential is

$$\Phi = \left(r + \frac{1}{r} \right) \cos \theta. \quad (50)$$

Its gradient gives the velocity of the flow. A well-known property is that the velocity is largest at the top and bottom of the cylinder ($r = 1, \theta = \pm\pi/2$), where its magnitude is twice that of the incident (or outgoing) flow far away, at infinity.

2.3. Poisson Integral

The Poisson Integral is a Green's function relation. Green's functions make it possible to express solutions to differential equations in integral form, relating the solution at a point to the solution on a boundary or in a region (depending on the context). Boundary could mean a time interval, in dynamical problems (when a point would be a given instant in time).

The Poisson Integral relates a harmonic function at a point to the values of the function on a circular boundary. Thus the Poisson Integral generates solutions to Laplace's equation when the boundary conditions supplied are those on a circle.

Consider an analytic function $f(z)$. Let C be a circle of radius a ; take the centre of the circle as the origin. Let z be a point inside C . Starting from the Cauchy integral formula:

$$\begin{aligned}
2\pi i f(z) &= \oint_C \frac{f(w)}{w-z} dw \\
&= \oint_C \frac{(w^* - z^*)f(w)}{(w-z)(w^* - z^*)} dw \\
&= \int_0^{2\pi} \frac{f(a, \phi) [a \exp(-i\phi) - r \exp(-i\theta)] ai \exp(i\phi)}{a^2 - 2ar \cos(\phi - \theta) + r^2} d\phi.
\end{aligned} \tag{51}$$

For the last line, we have written w and z in polar form:

$$w = a \exp(i\phi) \quad , \quad z = r \exp(i\theta) \tag{52}$$

with $r < a$ (as z is inside the circle). $f(a, \phi)$ denotes f on the circle in terms of the plane polar coordinates a, ϕ of w . Rearranging further:

$$\begin{aligned}
2\pi f(r, \theta) &= \int_0^{2\pi} \frac{(a^2 - r^2)f(a, \phi)}{a^2 - 2ar \cos(\phi - \theta) + r^2} d\phi \\
&+ \int_0^{2\pi} \frac{\{r^2 - ar \exp[i(\phi - \theta)]\} f(a, \phi)}{a^2 - 2ar \cos(\phi - \theta) + r^2} d\phi.
\end{aligned} \tag{53}$$

The last step is to show that the second integral above vanishes. We use the substitutions (52) to work our way backwards:

$$\begin{aligned}
&\int_0^{2\pi} \frac{\{r^2 - ar \exp[i(\phi - \theta)]\} f(a, \phi)}{a^2 - 2ar \cos(\phi - \theta) + r^2} d\phi \\
&= \oint_C \frac{z^* (z - w) f(w)}{(w - z)(w^* - z^*)} \frac{dw}{iw} \\
&= \oint_C \frac{iz^* f(w)}{a^2 - z^* w} dw \\
&= 0
\end{aligned} \tag{54}$$

since the only pole of the integrand, at $w = a^2/z^* = (a^2/r) \exp(i\theta)$, is outside C ($a^2/r > a$).

Therefore

$$f(r, \theta) = \frac{1}{2\pi} \int_0^{2\pi} \frac{(a^2 - r^2)f(a, \phi)}{a^2 - 2ar \cos(\phi - \theta) + r^2} d\phi. \tag{55}$$

This is the Poisson Integral, for f at a point (r, θ) inside a circular region, in terms of f on the boundary.

It applies individually to the harmonic functions u and v in $f = u + iv$ (by taking real and imaginary parts above).

2.4. Mean Value Property

Putting $r = 0$ above gives the relation

$$f(\text{centre of circle}) = \frac{1}{2\pi} \int_0^{2\pi} f(a, \phi) d\phi = \langle f(\text{boundary}) \rangle \quad (56)$$

between f at the centre of a circle and the average of f on the periphery of the circle. This relation applies to u and v individually.

3. Maximum Modulus Theorem

The Maximum Modulus Theorem is a close relative of the Mean Value Property and the Liouville Boundedness Theorem. Along with the “constant- f ” results, it reflects the internal structure that the definition of differentiability codes into complex functions.

It is a special case of the Cauchy inequalities. Let $f(z)$ be analytic in a region R , and let C be a circle of radius r centered at z_0 and lying wholly in R . Starting from the Cauchy integral formula:

$$\begin{aligned} f(z_0) &= \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz \\ \Rightarrow |f(z_0)| &= \left| \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz \right| \\ &\leq \frac{1}{2\pi} \oint_C \frac{|f(z)|}{r} |dz| \\ &\leq M \oint_C \frac{|dz|}{2\pi r} \end{aligned} \quad (57)$$

$$= M \Rightarrow |f(z_0)| \leq M \quad (58)$$

where M is the maximum of $|f(z)|$ on C . Therefore, the modulus of an analytic function at a point is always less than, or equal to, its maximum on a circle around the point.

It follows immediately that $|f(z)|$ cannot have a local maximum: if it did, then by considering a small circle around the maximum we would contradict (58).

Next, we show that equality above is achieved if and only if $f(z)$ is constant. Let $|f(z_0)| = M$. This requires, in particular, equality in step (57), which in turn requires $|f(z)| = M$ everywhere on C . Now consider a second, smaller circle C' , concentric with C . The maximum of $|f(z)|$ on C' , call it M' , must obey $M' \leq M$. But we must also have $|f(z_0)| \leq M'$. Hence, $M' = M$. But then $|f(z)| = M' = M$ everywhere on C' . Since C' is arbitrary, it follows that $|f(z)| = M = \text{const}$ everywhere in C . Hence, by 1.4, $f(z)$ is constant in C .

Now consider the region R , and let M be the maximum of $|f(z)|$ on the boundary of R . For any z_0 in R we must have

$$|f(z_0)| \leq M; \quad (59)$$

for otherwise $|f(z)|$ must have a local maximum in R , which is not possible. Suppose now we have equality above. Choose a circle C centered at z_0 and contained wholly in R . The maximum of $|f(z)|$ on C , M' , cannot exceed M , and at the same time we must have $|f(z_0)| \leq M'$. It follows

that $M' = M = |f(z_0)|$. By the earlier argument, $|f(z)| = M = |f(z_0)|$ on, and everywhere in, C . Hence, $f(z)$ is constant in C . But then all its derivatives vanish at any point in C . By considering the Taylor series for $f(z)$ about points in C we see, further, that $f(z)$ is constant in neighbourhoods that venture outside C . By paving R with overlapping neighbourhoods, we see that $f(z)$ is constant in R .

Therefore, if $f(z)$ is analytic in a region R , then either

- i. $f(z)$ is constant
- or
- ii. $|f(z)|$ attains its maximum on the boundary of R .

This is the Maximum Modulus Theorem.

4. Fundamental Theorem of Algebra

Let $P(z)$ be a non-constant polynomial of finite degree. Suppose it has no zeros, that is, it does not vanish for any finite z . Then

- i. $f(z) = 1/P(z)$ is entire (analytic everywhere)
- and
- ii. $\lim_{|z| \rightarrow \infty} |f(z)| = 0$.

Hence, by the Maximum Modulus Theorem, $|f(z)| = 0$ for all finite z , requiring $P(z)$ to be infinite for all z , which is nonsense. Therefore a non-constant polynomial must have at least one zero. This is the Fundamental Theorem of Algebra.

5. Integration

There is an amazing wealth of integrals that can be done by contour integration, and we have seen several types. Here are two interesting further examples.

i.

$$I = \int_0^\infty \frac{dx}{1+x^\alpha} \quad , \quad \alpha > 1$$

We will do the integral in a way that allows α to be non-integer (when, strictly speaking, a branch cut is needed since z^α is multivalued) and keeps the pole-counting to a minimum.

We restrict z to the section of the complex plane

$$0 \leq \arg z \leq \frac{2\pi}{\alpha} . \tag{60}$$

In this region $(1+z^\alpha)^{-1}$ has precisely one pole, at

$$z_0 = \exp(i\pi/\alpha) . \tag{61}$$

It is a simple pole with residue

$$\begin{aligned} r_0 &= \lim_{z \rightarrow z_0} \frac{z - z_0}{1 + z^\alpha} \\ &= \frac{1}{\alpha z_0^{\alpha-1}} \end{aligned} \tag{62}$$

using L'Hospital's rule. Thus,

$$r_0 = -\frac{z_0}{\alpha} = -\frac{\exp(i\pi/\alpha)}{\alpha}. \quad (63)$$

Consider the integral

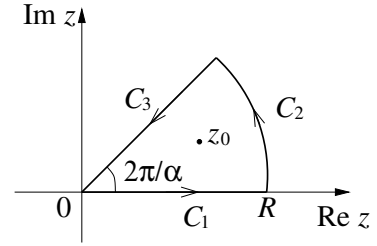
$$\oint_C \frac{dz}{1+z^\alpha} \quad (64)$$

around the boundary

$$C = C_1 + C_2 + C_3 \quad (65)$$

where

- C_1 runs from $z = 0$ to $z = R$ along the real axis;
- C_2 is the circular arc $z = R \exp(i\theta)$, from $\theta = 0$ to $\theta = 2\pi/\alpha$;
- C_3 runs back to the origin along the ray $z = r \exp(2\pi i/\alpha)$, from $r = R$ to $r = 0$.



Now take the limit $R \rightarrow \infty$. The contribution from C_2 vanishes by the Darboux inequality (provided $\alpha > 1$). Consider the contribution from C_3 :

$$\begin{aligned} \int_{C_3} \frac{dz}{1+z^\alpha} &= \int_\infty^0 \frac{dr \exp(2\pi i/\alpha)}{1+r^\alpha} \\ &= -\exp(2\pi i/\alpha) \int_0^\infty \frac{dr}{1+r^\alpha} \\ &= -\exp(2\pi i/\alpha) I. \end{aligned} \quad (66)$$

Therefore, by the residue theorem:

$$[1 - \exp(2\pi i/\alpha)] I = 2\pi i r_0 = -2\pi i \frac{\exp(i\pi/\alpha)}{\alpha} \quad (67)$$

and hence

$$\int_0^\infty \frac{dx}{1+x^\alpha} = \frac{\pi/\alpha}{\sin(\pi/\alpha)}. \quad (68)$$

When $\alpha \rightarrow 1$, the integral grows without limit. When $\alpha \rightarrow \infty$, the integral tends to 1: the integrand then is unity for $0 < x < 1$ and zero for $x > 1$.

ii.

$$\int_0^\infty \cos(x^2) dx \quad \text{and} \quad \int_0^\infty \sin(x^2) dx$$

We consider

$$\oint_C \exp(iz^2) dz \tag{69}$$

with a contour C composed of three parts:

$$C = C_1 + C_2 + C_3 \tag{70}$$

where

- C_1 runs from $z = 0$ to $z = R$ along the real axis;
- C_2 is the circular arc $z = R \exp(i\theta)$, from $\theta = 0$ to $\theta = \pi/4$;
- C_3 runs back to the origin along the ray $z = r \exp(i\pi/4)$, from $r = R$ to $r = 0$.

In the limit $R \rightarrow \infty$, the contribution from C_2 vanishes (by a variation on Jordan's Lemma). The integral on C_3 becomes

$$\begin{aligned} \int_{C_3} \exp(iz^2) dz &= \int_\infty^0 \exp(i\pi/4) dr \exp(-r^2) \\ &= -\frac{1+i}{\sqrt{2}} \frac{\sqrt{\pi}}{2}. \end{aligned} \tag{71}$$

Finally, (69) vanishes, since the integrand is entire. Therefore

$$\int_0^\infty \exp(ix^2) dx = \frac{1+i}{\sqrt{2}} \frac{\sqrt{\pi}}{2} \tag{72}$$

and hence

$$\int_0^\infty \cos(x^2) dx = \int_0^\infty \sin(x^2) dx = \frac{1}{2} \sqrt{\frac{\pi}{2}}. \tag{73}$$

The integrands, $\cos(x^2)$ and $\sin(x^2)$, oscillate without limit as $x \rightarrow \infty$, but the integrals exist: the oscillations in the integrands speed up sufficiently quickly as x grows, to yield cancellations efficient enough to make the integrals converge.

Tchavdar Todorov

with thanks to Gleb Gribakin
for invaluable suggestions